Computer Science &

Data Science

ware housing

Warehousing

Lecture -09

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Recap of Previous Lecture

Topic Sampling

Topic Types of Sampling

Topic Compression

Topics to be Covered

Topic Huffman Encoding

Topic Data Warehouse Measures

Lempel–Ziv–Welch Algorithm

Example:

AABABABBABABBBAABAB

Position	A	AB	ABA	B	BA	BAB	BB	AA	BAB
Sequence	1	2	3	4	5	6	7	8	9
Numerical Presentation	ϕA	1B	2A	ϕB	4A	5B	4B	1A	5B
Code	0000	0011	0100	0001	1000	1011	1001	0010	1011

A → 0

B → 1

Topic : Huffman Encoding

Assigns variable-length codes to input characters, lengths of the assigned codes are based on the frequencies of corresponding characters.

smaller code to frequent characters

There are mainly two major parts in Huffman Coding

1. Build a Huffman Tree from input characters.
2. Traverse the Huffman Tree and assign codes to characters.

→ left 0
→ right 1

Topic : Huffman Encoding

Assume a message having 50 characters total with

A: 20 times

B: 15 times

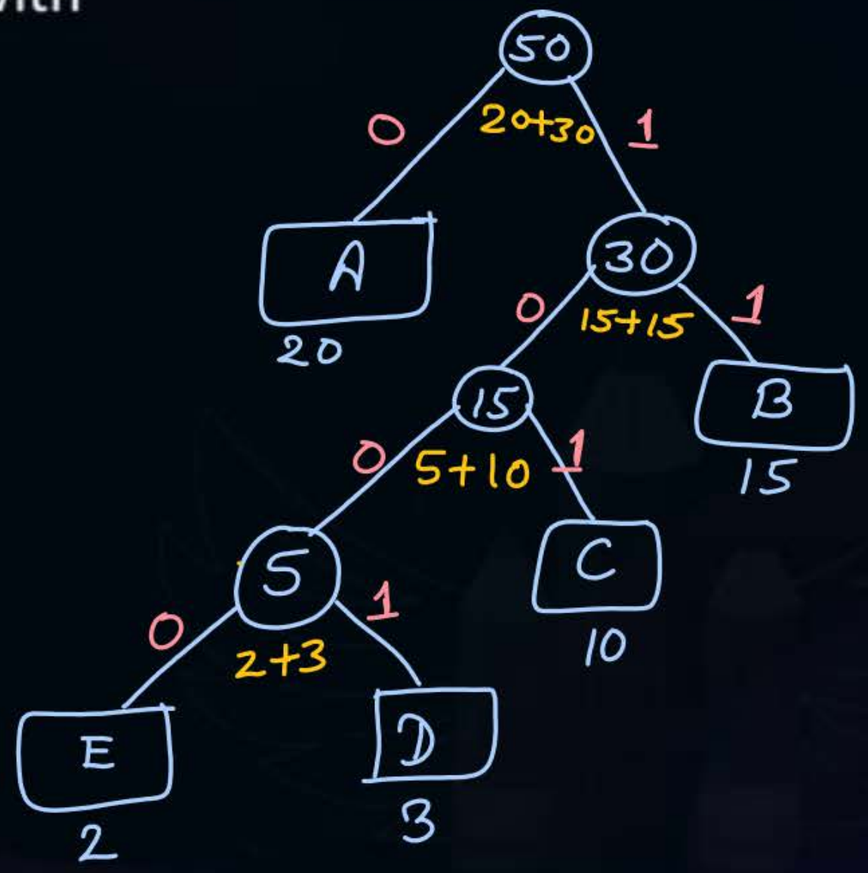
C: 10 times

D: 3 times

E: 2 times

start with 2 smallest freq. characters

small freq in left side



	Code
A	0
B	11
C	101
D	1001
E	1000

How to write BAD in encoded form

11 0 1001

Length of encoded message:-

A: 20 * 1 bits = 20

B: 15 * 2 bits = 30

C: 10 * 3 bits = 30

D: 3 * 4 bits = 12

E: 2 * 4 bits = 8

100 bits

Topic : Huffman Encoding

Size of encoded message = $\sum d_i * fi$

d_i = size of encoded character i

f_i = frequence of character i in message

Topic : Huffman Encoding

Assume a message having 100 characters total with

A: 10 times ⇒ 0.1 ⇒ 10%

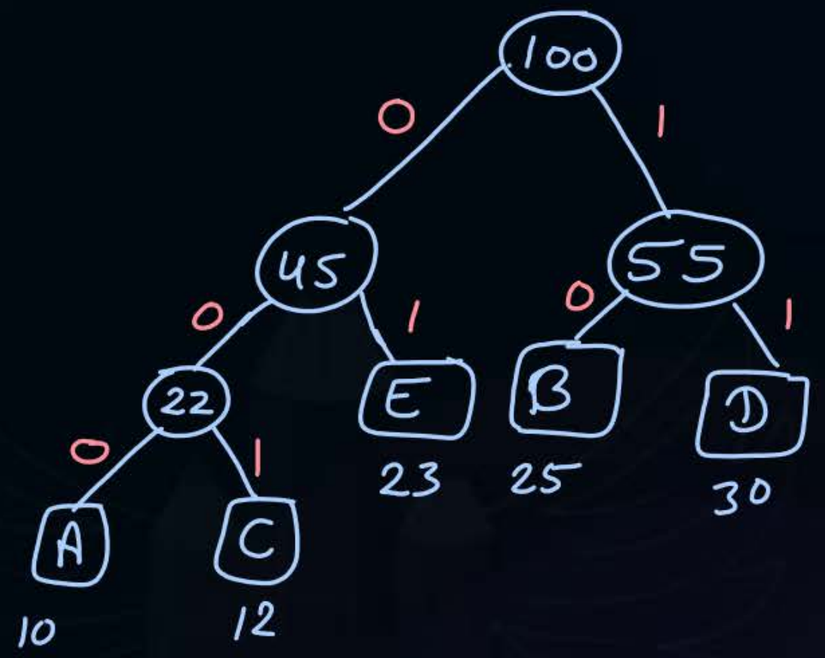
B: 25 times ⇒ 0.25 ⇒ 25%

C: 12 times ⇒ 0.12 ⇒ 12%

D: 30 times ⇒ 0.3 ⇒ 30%

E: 23 times

	Encoding
A	000
B	10
C	001
D	11
E	01



The number of bits sent for a message: "BADACADEE"

$\underbrace{10\ 000\ 11\ 000\ 001\ 000\ 11\ 01\ 01}_{22 \text{ bits}}$

[MCQ]

#Q. Huffman tree is constructed for the following data: {A, B, C, D, E}, with frequencies {0.17, 0.11, 0.24, 0.33, 0.15} respectively. 100 00 01101 is decoded as?

B A C E

(A) BACE

(B) CADE

(C) BAD

(D) CADD

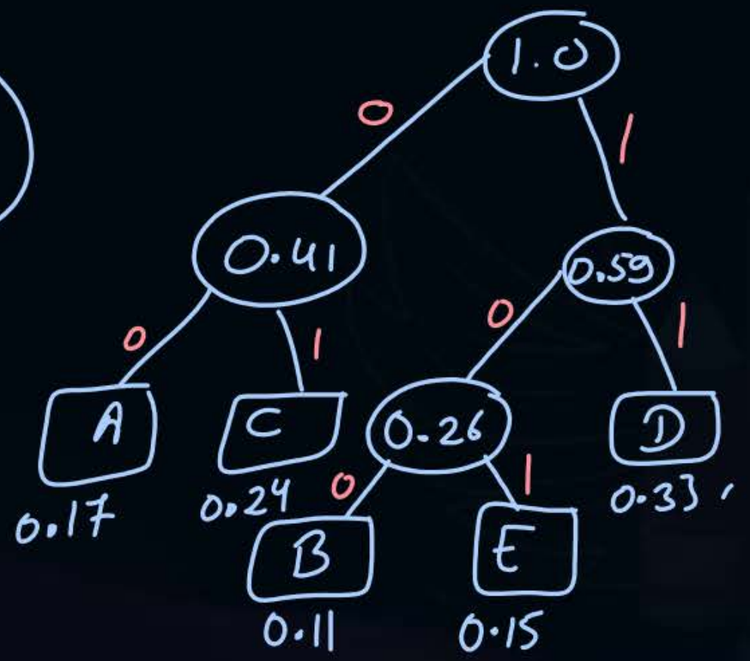
A: 0.17

B: 0.11

C: 0.24

D: 0.33

E: 0.15



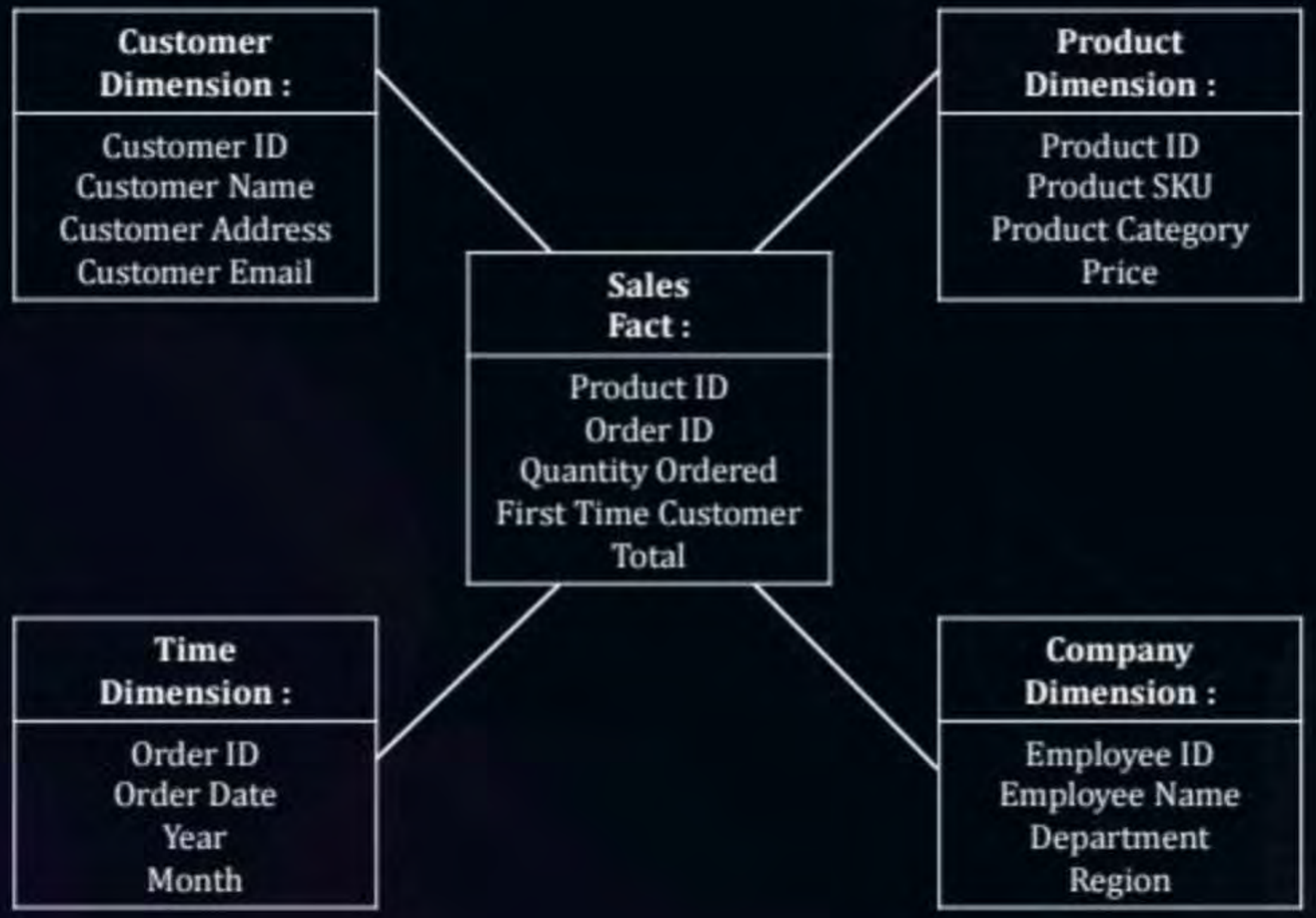
Topic : Datawarehouse Measures

In a data warehouse, a measure is a property on which calculations (e.g., sum, count, average, minimum, maximum) can be made.

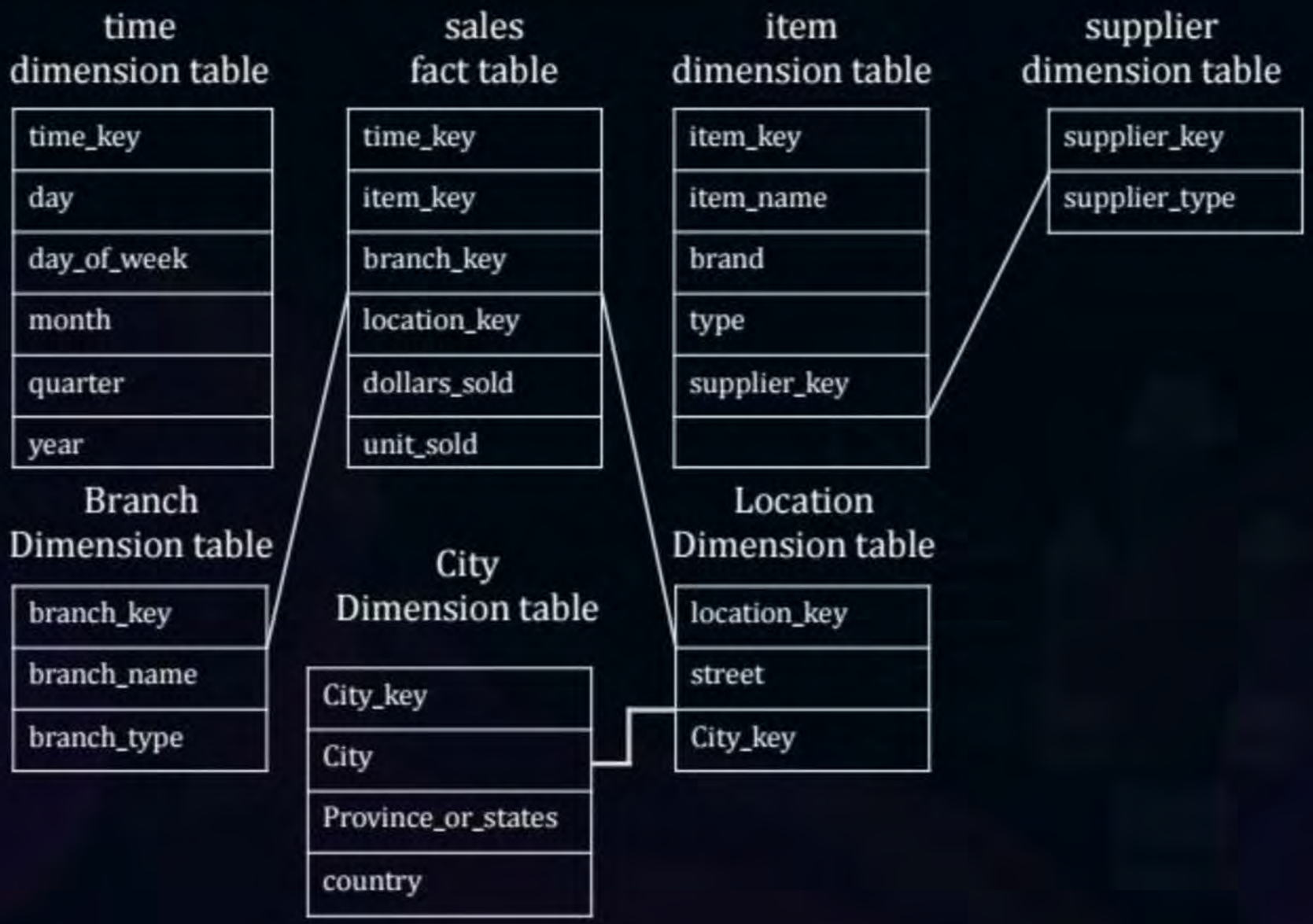
Topic : Datawarehouse Schema

1. Star Schema
2. Snowflake Schema
3. Fact Constellation Schema

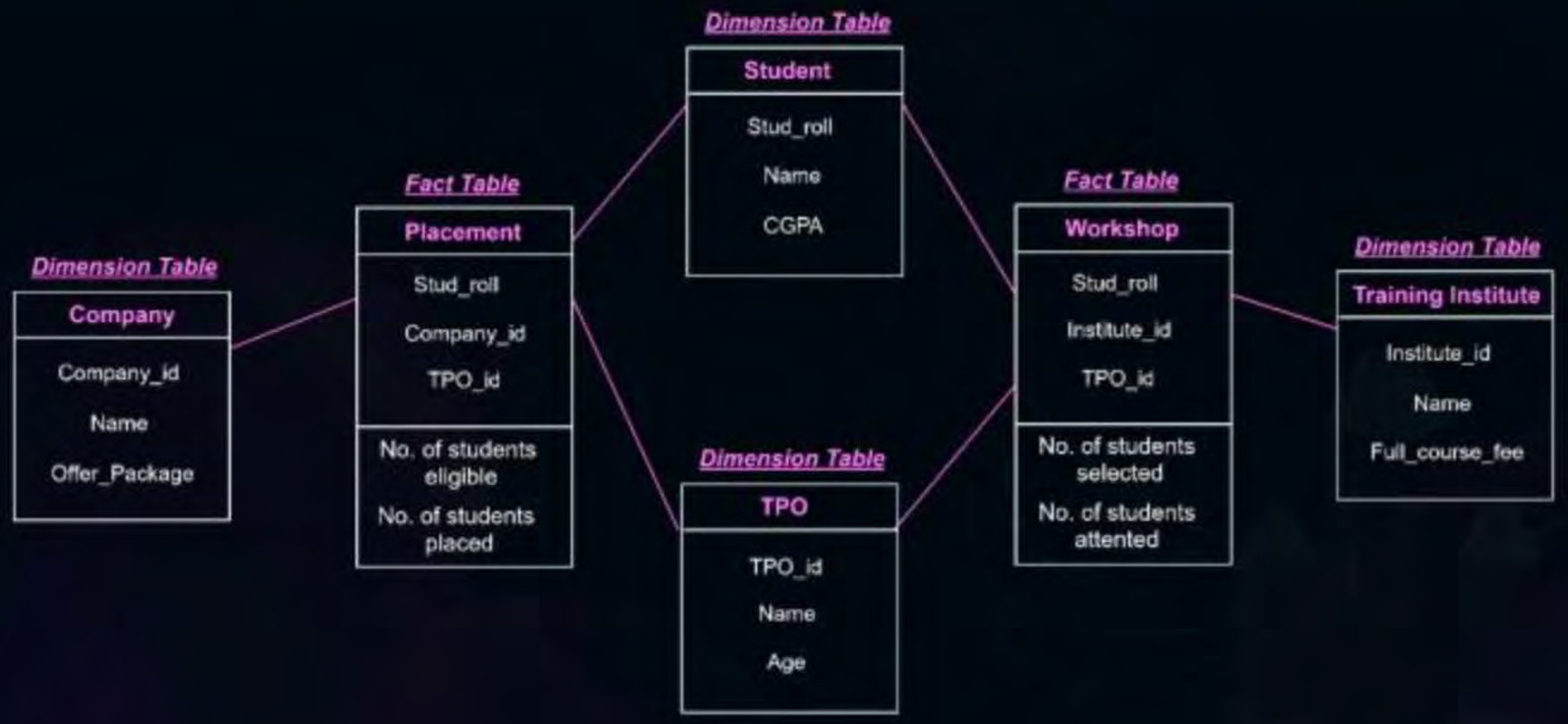
Topic : Star Schema



Topic : Snowflake Schema



Topic : Fact Constellation Schema (Galaxy Schema)



Topic : Datawarehouse Measures

In a data warehouse, a measure is a property on which calculations (e.g., sum, count, average, minimum, maximum) can be made.

For example, if a retail store sold a specific product, the quantity and prices of each item sold could be added or averaged to find the total number of items sold or the total or average price of the goods sold.

Topic : Datawarehouse Measures

Dimensions:

- The cap was sold
- Color of the cap
- Size of the cap
- The name of the company
- Employee's name
- An employee's office location
- A warehouse where a cap is kept for sale

Topic : Datawarehouse Measures

Facts:

- Selling a cap for ₹250 on the internet
- The existing warehouse receives a shipment of 100 caps from a supplier
- A consumer is receiving two caps
- Getting a ₹ 200 late payment from a customer
- An employee is paid a salary of ₹15,00,00 per year

Topic : Datawarehouse Measures

Measures:

- It was sold at 1 p.m. the day before yesterday
- It was purchased for ₹350
- A consumer is receiving two hats
- SFAF3423 is the product ID

Products.

Dimension: shops

shopid
shopname
location

Fact: sales

shopid
pid
Eid
cid
sale amount

Dimension: Employee

Eid
Ename
dob
phone
location

Dimension: customers

cid
cname
dob
;

Topic : Datawarehouse Measures: Types

1. Additive
2. Semi additive
3. Non-Additive

Topic : Datawarehouse Measures: Additive

Additive (summarize) across all dimensions associated with fact table

Shop Id	Employee Id	Customer Id	Sales
S1	E1	C1	16000
S1	E2	C1	10000
S2	E1	C2	5000
S3	E3	C3	25000
S3	E3	C2	20000

ex:- sum of sales for a specific shopid or customerid or employee id

Topic : Datawarehouse Measures: Semi-Additive

Additive (summarize) across some dimensions associated with fact table

closing

Date Key	Customer Id	Account Number	Account _ Balance
1-Dec-2023	C1	1	1000
2-Dec-2023	C1	1	1200
3-Dec-2023	C1	1	1200
4-Dec-2023	C1	1	1100
1-Dec-2023	C1	2	2000
2-Dec-2023	C1	2	2000
3-Dec-2023	C1	2	2000
4-Dec-2023	C1	2	2100

saving

deposit (FD)

3200

Ex:- Total balance of a customer with id C1 on a given date

	Ac no	
3-Dec	1	1200
3-Dec	2	2000
		3200

3200 → semi-additive measure

Topic : Datawarehouse Measures: Non - Additive

Additive (summarize) across no any dimensions associated with fact table

Topic : Datawarehouse Measures: Non - Additive

Example: Ratio and percentage attributes

Quarter	Org Id	Gross Profit Margin
2021Q1	1	20%
2021Q2	1	19%
2021Q3	1	29%
2021Q4	1	21%
2021Q1	2	5%
2021Q2	2	17%
2021Q3	2	12%
2021Q4	2	36%

$$avg = \frac{20+19+29+21}{4}$$

but addition not working here.

2 mins Summary

Topic Huffman Encoding

Topic Data Warehouse Measures

Happy Learning

THANK - YOU